Gram-Schmidt

Def^n $W \leq \mathbb{R}^n$

An orthonormal basis for W is a basis $(u_1, \ldots, u_m)$ for W with the following properties:

1. $u_i \cdot u_j = 0$ if $i \neq j$
2. $\underbrace{u_i \cdot u_i = 1}$ $\forall i$

$$u_i = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \qquad \Updownarrow \quad a_1^2 + a_2^2 + \ldots + a_n^2 = 1$$

Thm (Gram-Schmidt)

(1) Every $W \leq \mathbb{R}^n$ has an orthonormal basis

(2) In fact, suppose that we have a basis

$(v_1, v_2, .., v_m)$

for W

Then we can find an orthonormal basis $(u_1, u_2, .., u_m)$ for W and an invertible upper triangular matrix

$R \in M_{m \times m}(\mathbb{R})$ s.t.

$$\underbrace{[v_1 \ldots v_m]}_{} = \underbrace{[u_1 \ldots u_m]}_{} \underbrace{R}_{}$$

$n \times m$ $\qquad\qquad\qquad$ $n \times m$ $\quad$ $n \times m$

Rmk ① $Q = \underbrace{[u_1\ u_2 \cdots u_m]}_{n \times m}$

$$\underbrace{Q^T}_{n \times n} \underbrace{Q}_{n \times m} = \begin{bmatrix} u_1^T \\ u_2^T \\ \vdots \\ u_m^T \end{bmatrix} [u_1\ u_2 \cdots u_m]$$

$$= \begin{bmatrix} \overset{1}{u_1 \cdot u_1} & \overset{0}{u_1 \cdot u_2} & \cdots & \overset{0}{u_1 \cdot u_m} \\ \overset{0}{u_2 \cdot u_1} & \overset{1}{u_2 \cdot u_2} & & \vdots \\ \vdots & & \ddots & \\ & & \cdots & \overset{1}{u_m \cdot u_m} \end{bmatrix}$$

$$= I_m$$

In particular, if $n = m$ (i.e. $W = \mathbb{R}^n$)

then $Q \in M_{n \times n}(\mathbb{R})$ is invertible

& $Q^{-1} = Q^T$

Def$^{\underline{n}}$ $Q \in M_{n\times n}(\mathbb{R})$ is

orthogonal if one of

the following equivalent conditions

holds:

1. $Q^T = Q^{-1}$
2. $Q^T Q = Q Q^T = I_n$
3. The columns of Q form an orthonormal basis for $\mathbb{R}^n$
4. The rows of Q form an orthonormal basis for $\mathbb{R}^n$

The Gram-Schmidt algorithm

(1) Start with m linearly ind. vectors

$$v_1, v_2, \ldots, v_m \in \mathbb{R}^n$$

Will produce an orthonormal list of vectors by induction on m.

(2) (Base case: $m = 1$)

$$0 \neq v_1 \in \mathbb{R}^n$$

Def^n $v \in \mathbb{R}^n$, $\underbrace{\|v\|}_{\text{"norm of } v\text{"}} = \sqrt{v \cdot v}$

$v = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$, $\|v\| = \sqrt{a_1^2 + a_2^2 + \ldots + a_n^2}$

<u>Observation:</u> (1) $v \in \mathbb{R}^n$, $\|v\| = 0$

$\iff v = 0$

This property is the key reason to restrict to $\mathbb{F} = \mathbb{R}$

(2) $0 \neq v \in \mathbb{R}^n$, $c \in \mathbb{R}$, $\|c \cdot v\| = c \cdot \|v\|$

$$\left\| \frac{v}{\|v\|} \right\| = \left\| \frac{1}{\|v\|} \cdot v \right\|$$

$$= \frac{1}{\|v\|} \|v\| = 1$$

<u>Back to base case:</u>

$0 \neq v_1 \in \mathbb{R} \rightsquigarrow u_1 = \dfrac{v_1}{\|v_1\|}$ will be orthonormal

(3) (Inductive hypothesis)

$m \geq 2$, $(v_1, .., v_{m-1})$: linearly independent

Then we can find: (a) $(u_1, .., u_{m-1})$ orthonormal

(b) $R' \in M_{(n-1)\times(n-1)}(\mathbb{R})$

invertible upper triangular

s.t.

$$[v_1 \ldots v_{m-1}] = [u_1 \ldots u_{m-1}] R'$$

Let's break down what this means

$$\underbrace{[u_1\, u_2 \ldots u_{m-1}]}_{Q'} \begin{bmatrix} a_{1,1} & a_{1,2} & - & - a_{1,m-1} \\ 0 & a_{2,2} & & \vdots \\ 0 & 0 & \ddots & \vdots \\ \vdots & \vdots & & \vdots \\ 0 & 0 & & a_{m-1,m-1} \end{bmatrix}$$

$$= \begin{bmatrix} a_{1,1}u_1 & a_{1,2}u_1 + a_{2,2}u_2 & \ldots & \sum_{i=1}^{m-1} a_{i,m-1} u_i \end{bmatrix}$$

- $Q'R'e_1 = Q'(a_{1,1} e_1)$

$= a_{1,1} Q'e_1$

$= a_{1,1} u_1$

- $Q'R'e_2 = Q'(a_{1,2} e_1 + a_{2,2} e_2)$

$= a_{1,2} u_1 + a_{2,2} u_2$

We're saying:

$v_1 = a_{1,1} u_1$

non-zero

$v_2 = a_{1,2} u_1 + a_{2,2} u_2$

$\vdots$

$v_{m-1} = a_{1,m-1} u_1 + \dots + a_{m-1,m-1} u_m$

④ (Inductive step)

$(v_1, \ldots, v_{m-1}, v_m)$: linearly independent

$\Downarrow$ want

① $(u_1, \ldots, u_{m-1}, u_m)$: orthonormal

② $$R = \begin{array}{cc} & \overbrace{}^{m-1} \\ m-1\left\{\right. & \begin{bmatrix} R' & * \\ 0 & a_{m,m} \end{bmatrix} \end{array} \in M_{m\times m}(\mathbb{R})$$

s.t. $[v_1 \ldots v_m] = [u_1, \ldots, u_m] R$.

Consider the simplest case where

$$(v_1, \ldots, v_{m-1}) = (u_1, \ldots, u_{m-1})$$

so that $R' = I_{m-1}$

want:

$$[u_1, \ldots u_{m-1}, v_m] = [u_1, \ldots, u_m] \begin{bmatrix} I_{m-1} & * \\ 0 & a_{m,m} \end{bmatrix}$$

want:

$$V_m = a_{1,m} u_1 + a_{2,m} u_2 + \dots + a_{m-1,m} u_{m-1} + \underbrace{a_{m,m}}_{\neq 0} u_m$$

$$\Leftrightarrow \quad u_m = b_{1,m} u_1 + b_{2,m} u_2 + \dots + b_{m-1,m} u_{m-1} + b_{m,m} V_m$$

s.t. (1) $u_m \cdot u_i = 0 \quad \forall i < m$

(2) $u_m \cdot u_m = 1$

For (1): $u_m \cdot u_i \quad \forall i < m$

$$= (b_{1,m} u_1 + \dots + b_{m-1,m} u_{m-1} + b_{m,m} V_m) \cdot u_1$$

$$= b_{1,m}(u_1 \cdot u_1) + b_{2,m}(u_2 \cdot u_1) + \dots + b_{m-1,m}(u_{m-1} \cdot u_1) + b_{m,m}(V_m \cdot u_1)$$

$$= b_{i,m} + b_{m,m}(V_m \cdot u_i) = 0$$

$$\Rightarrow \quad b_{i,m} = -b_{n,m}\,(v_n \cdot u_i)$$